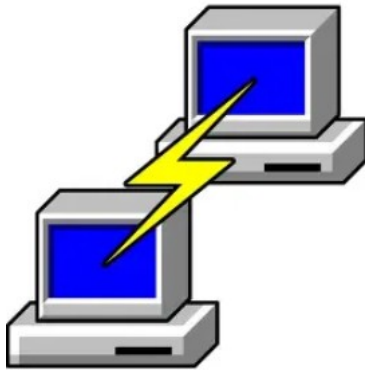




MULTI AREA OSPF CONFIGURATION

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Purpose:

To Refresh OSPF concepts and multi-layer switches as well as to test out the new hardware and verify its functionality.

Background:

OSPF (Open Shortest Path First) is a routing protocol developed in 1989 for TCP (Transmission Control Protocol)/IP (Internet Protocol) networks. OSPF supports both IPv4 and IPv6 networks. It is used in many large-scale enterprise networks to this day. OSPF achieves this by creating multiple areas in the single OSPF network. While OSPF can be used in large scale networks, it can also be used in small SOHO or private networks by using a single area OSPF network. Rather than statically setting every route on the network, you just need to advertise the IP addresses on each router, making it much quicker. If you have an enterprise network with multi area OSPF and you are adding a new router, you could just use one command to advertise the network on OSPF, but if you used static routes, you would have to statically set up a new route for every router.

Before OSPF was used RIP was the standard for dynamic routing. RIP was very simple to use for many people and was easy to understand. RIP was better than previous routing protocols and was able to be used and works consistently on smaller networks. While it was better than past protocols RIP had several major problems. RIP struggled with larger area networks because it didn't always choose the best and fastest route. RIP choose the route with

the least hops on the network rather than choosing the fastest route with faster links even if it was one or two hops more. It also struggled in larger scale networks because it was not always able to adapt to change on the network. OSPF was much preferred to RIP after its release because it fixed most of the major problems RIP had. OSPF choose the route that had the fastest connection even if it involved choosing the route with more hops. OSPF was also able to adapt to change on networks making it a viable option for some larger scale networks.

OSPF was updated to OSPFv2 in 1991. OSPFv2 added areas into OSPF and added support for multi area OSPF which further allowed for larger scale networks. OSPFv2 also made area 0 the backbone area for all networks. OSPFv2 was specifically made for IPv4 as opposed to OSPFv1. OSPF was updated to OSPFv3 in 2008. OSPFv3 mainly focused on IPV6. It allowed OSPF to use the newer IPv6 addresses that were 128-bit. The IPv4 address pool was getting smaller and needed to have more address added. IPv6 solved this problem, but protocols still needed to add compatibility. OSPFv3 added this support for IPv6. While OSPFv3 did add several other minor changes and security updates, it is mostly known for its ipv6 support.

OSPF can be configured as single area, which is commonly used in smaller networks. Single area OSPF has a simpler configuration, while also being relatively easy to troubleshoot for issues. While single area OSPF works well for some cases in smaller networks it is limited by scalability. Multi area OSPF can be scaled much more than single area and can be used in large-scale enterprise networks. Multi area networks can also be more flexible because topology changes do not affect the entire network, just the area that the change occurred in. Routers do not have to advertise or learn information from every router in the network but just the routers in their area.

In our topology we used two multi-layer switches. A multi-layer switch is designed to act as a regular switch (Operating in layer 2) having the regular functionality of a layer 2 switch with capabilities such as VLANs and spanning tree but also having the capabilities of a router (Operating in layer 3). They become especially helpful for inter-vlan routing. Inter-vlan routing allows you to virtually create separate routes on one multi-layer switch.

Lab summary:

We built the topology for the network containing five routers and 2 multi-layer switches. We then configured ipv4 addresses and Multi-area OSPF on the routers and switches. After the configurations were complete, we did the verification show commands along with pings and traceroutes. After troubleshooting the network, we were able to traceroute through the network. We then set up ipv6 on our network and configured OSPFv3 on the devices. We were able to traceroute through the network with ipv6.

Lab commands:

Area 1:

Switch1:

Switch>

Switch>en

Switch#conf t

Switch(config)#ip routing`

Switch(config)#ipv6 unicast-routing

Switch(config)#int g1/0/1

Switch(config-if)#no switchport

Switch(config-if)#ip add 192.168.1.1 255.255.255.252

Switch(config-if)#ipv6 add 2001:db8:acad:1::1/126

Switch(config-if)#no shut

Switch(config-if)#ip ospf 6 area 1

Switch(config-if)#ipv6 ospf 6 area 1

Switch(config-if)#router ospf 6

Switch(config-router)#network 192.168.1.0 255.255.255.252 area 1

Switch(config-router)#router-id 6.6.6.6

Switch(config-router)#do clear ip ospf process

Switch(config-router)#exit

Switch(config-router)#router ospfv3 6

Switch(config-router)#router-id 6.6.6.6

Switch(config-router)#exit

Switch(config)#hostname SWITCH1

Router 1:

Router#en

Router#conf t

Router(config)#ipv6 unicast-routing

Router(config)# ip routing

Router(config)#int g0/0/0

Router(config-if)#ip address 192.168.1.2 255.255.255.252

Router(config-if)#ipv6 address 2001:db8:acad:1::2 /126

Router(config-if)#no shut

Router(config-if)#ip ospf 1 area 1

Router(config-if)#ipv6 ospf 1 area 1

Router(config-if)#router ospf 1

Router(config-router)#router-id 1.1.1.1

Router(config-router)#exit

Router(config)#int g0/0/1

```
Router(config-if)#ip address 192.168.1.5 255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:1::5/126
Router(config-if)#no shut
Router(config-if)#ip ospf 1 area 1
Router(config-if)#ipv6 ospf 1 area 1
Router(config-if)#router ospf 1
Router(config-router)#router-id 1.1.1.1
Router(config-router)#exit
Router(config)#hostname Router1
```

Router 2:

```
Router#en
Router#conf t
Router(config)#ipv6 unicast-routing
Router(config)#ip routing
Router(config)#int g0/0/0
Router(config-if)#ip address 192.168.1.6 255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:1::6/126
Router(config-if)#no shut
Router(config-if)#ip ospf 2 area 1
Router(config-if)#ipv6 ospf 2 area 1
Router(config-if)#router ospf 2
Router(config-router)#router-id 2.2.2.2
Router(config-router)#exit
Router(config)#int g0/0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:2::1/126
Router(config-if)#no shut
Router(config-if)#ip ospf 2 area 0
Router(config-if)#ipv6 ospf 2 area 0
Router(config-if)#router ospf 2
Router(config-router)#router-id 2.2.2.2
Router(config-router)#exit
Router(config)#hostname Router2
```

Area 0:

Router 3:

```
Router#en
Router#conf t
Router(config)#ipv6 unicast-routing
Router(config)#ip routing
Router(config)#int g0/0/0
Router(config-if)#ip address 192.168.2.2 255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:2::2/126
Router(config-if)#no shut
Router(config-if)#ip ospf 3 area 0
Router(config-if)#ipv6 ospf 3 area 0
```

```
Router(config-if)#router ospf 3
Router(config-router)#router-id 3.3.3.3
Router(config-router)#exit
Router(config)#int g0/0/1
Router(config-if)#ip address 192.168.2.5 255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:2::5/126
Router(config-if)#no shut
Router(config-if)#ip ospf 3 area 0
Router(config-if)#ipv6 ospf 3 area 0
Router(config-if)#router ospf 3
Router(config-router)#router-id 3.3.3.3
Router(config-router)#exit
Router(config)#hostname Router3
```

Area 2:

Router 4:

Router#en

Router#conf t

Router(config)#ipv6 unicast-routing

Router(config)#ip routing

Router(config)#int g0/0/0

Router(config-if)#ip address 192.168.2.6 255.255.255.252

Router(config-if)#ipv6 address 2001:db8:acad:2::6/126

Router(config-if)#no shut

Router(config-if)#ip ospf 4 area 0

Router(config-if)#ipv6 ospf 4 area 0

Router(config-router)#exit

Router(config)#int g0/0/1

Router(config-if)#ip address 192.168.3.6 255.255.255.252

Router(config-if)#ipv6 address 2001:db8:acad:3::6/126

Router(config-if)#no shut

Router(config-if)#ip ospf 4 area 2

Router(config-if)#ipv6 ospf 4 area 2

Router(config-if)#router ospf 4

Router(config-router)#router-id 4.4.4.4

Router(config-router)#network 192.168.2.4 0.0.0.3 area 0

Router(config-router)#network 192.168.3.4 0.0.0.3 area 2

Router(config-router)#exit

Router(config)#hostname Router4

Router 5:

Router#en

Router#conf t

Router(config)#ipv6 unicast-routing

Router(config)#ip routing

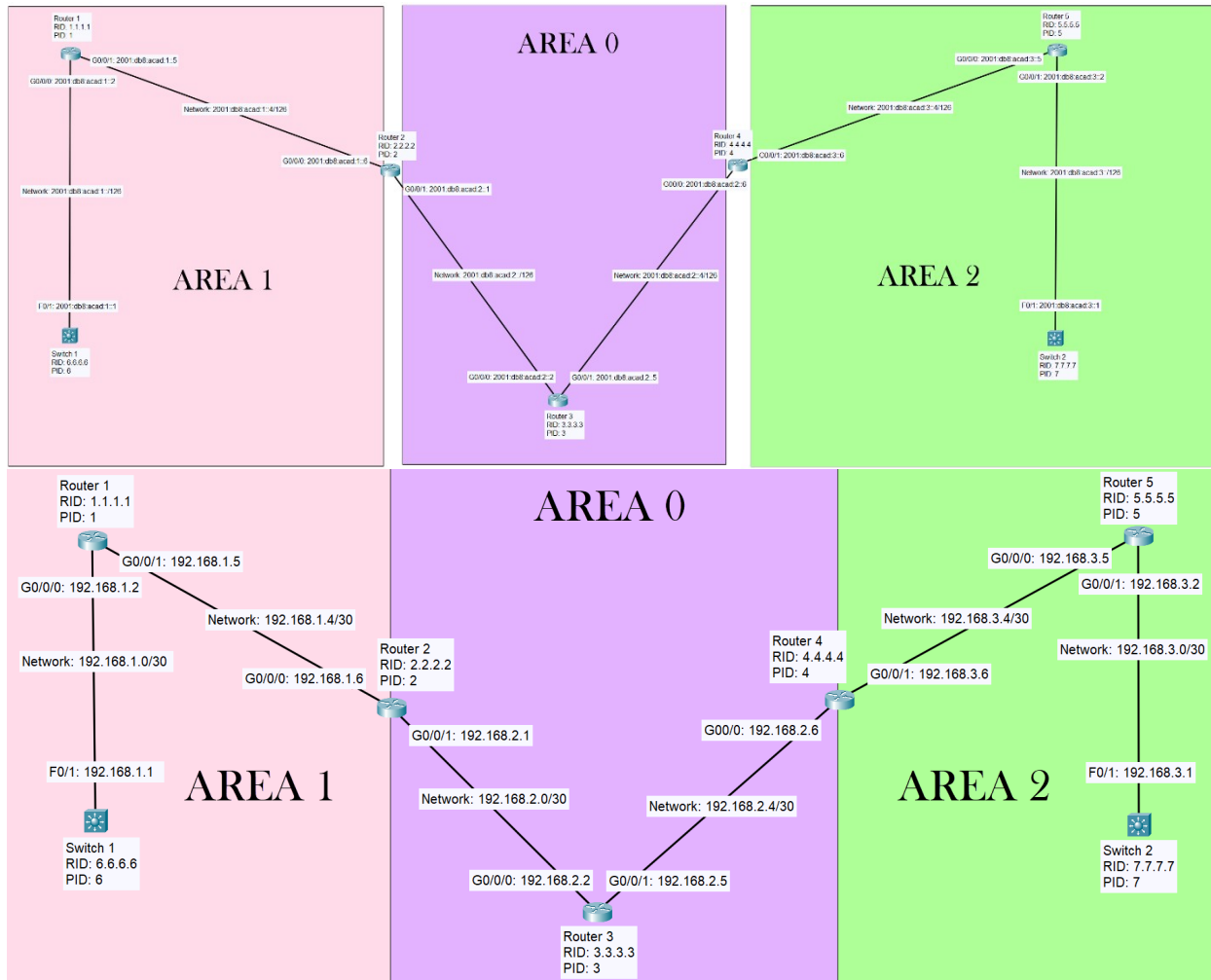
Router(config)#int g0/0/0

Router(config-if)#ip address 192.168.3.5 255.255.255.252

```
Router(config-if)#ipv6 address 2001:db8:acad:3::5/126
Router(config-if)#no shut
Router(config-if)#ip ospf 5 area 2
Router(config-if)#ipv6 ospf 5 area 2
Router(config-if)#router ospf 5
Router(config-router)#router-id 5.5.5.5
Router(config-router)#exit
Router(config)#int g0/0/1
Router(config-if)#ip address 192.168.3.2 255.255.255.252
Router(config-if)#ipv6 address 2001:db8:acad:3::2/126
Router(config-if)#no shut
Router(config-if)#ip ospf 5 area 2
Router(config-if)#ipv6 ospf 5 area 2
Router(config-if)#router ospf 5
Router(config-router)#router-id 5.5.5.5
Router(config-router)#exit
Router(config)#hostname Router5
```

```
Switch 2:
Switch#en
Switch#conf t
Switch(config)#ip routing
Switch(config)#ipv6 unicast-routing
Switch(config)#int g1/0/1
Switch(config-if)#no switchport
Switch(config-if)#ip address 192.168.3.1 255.255.255.252
Switch(config-if)#no shut
Router(config-if)#ipv6 address 2001:db8:acad:3::1/126
Switch(config-if)#ip ospf 7 area 2
Switch(config-if)#router ospf 7
Switch(config-router)#router-id 7.7.7.7
Switch(config-router)#exit
Switch(config)#int g1/0/1
Switch(config-if)#ipv6 ospf 7 area 2
Switch(config-if)#do clear ip ospf process
Switch(config-if)#exit
Switch(config)#hostname SWITCH2
```

Network Diagram:



Configs:

Area 1:

Switch 1:

```
Switch1(config)#do traceroute 192.168.3.1
```

Type escape sequence to abort.

Tracing the route to 192.168.3.1

VRF info: (vrf in name/id, vrf out name/id)

```
1 192.168.1.2 0 msec 4 msec 0 msec
2 192.168.1.6 0 msec 4 msec 0 msec
3 192.168.2.2 0 msec 4 msec 0 msec
4 192.168.2.6 0 msec 4 msec 0 msec
5 192.168.3.5 4 msec 4 msec 0 msec
6 192.168.3.1 4 msec 4 msec *
```

```
Switch1(config)#do traceroute 2001:db8:acad:3::1
```

```
Type escape sequence to abort.
```

```
Tracing the route to 2001:DB8:ACAD:3::1
```

```
1 2001:DB8:ACAD:1::2 0 msec 0 msec 4 msec
2 2001:DB8:ACAD:1::6 0 msec 0 msec 4 msec
3 2001:DB8:ACAD:2::2 0 msec 4 msec 0 msec
4 2001:DB8:ACAD:2::6 0 msec 4 msec 0 msec
5 2001:DB8:ACAD:3::5 0 msec 4 msec 4 msec
6 2001:DB8:ACAD:3::1 0 msec 4 msec 0 msec
```

```
Switch1(config)#do show ip ospf
```

```
Routing Process "ospf 6" with ID 192.168.1.1
```

```
Start time: 00:04:43.184, Time elapsed: 2d21h
```

```
Supports only single TOS(TOS0) routes
```

```
Supports opaque LSA
```

```
Supports Link-local Signaling (LLS)
```

```
Supports area transit capability
```

```
Supports NSSA (compatible with RFC 3101)
```

```
Supports Database Exchange Summary List Optimization (RFC 5243)
```

```
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
```

```
Router is not originating router-LSAs with maximum metric
```

```
Initial SPF schedule delay 50 msecs
```

```
Minimum hold time between two consecutive SPFs 200 msecs
```

```
Maximum wait time between two consecutive SPFs 5000 msecs
```

```
Incremental-SPF disabled
```

```
Initial LSA throttle delay 50 msecs
```

```
Minimum hold time for LSA throttle 200 msecs
```

```
Maximum wait time for LSA throttle 5000 msecs
```

```
Minimum LSA arrival 100 msecs
```

```
LSA group pacing timer 240 secs
```

```
Interface flood pacing timer 33 msecs
```

```
Retransmission pacing timer 66 msecs
```

```
EXCHANGE/LOADING adjacency limit: initial 300, process maximum 300
```

```
Number of external LSA 0. Checksum Sum 0x000000
```

```
Number of opaque AS LSA 0. Checksum Sum 0x000000
```

```
Number of DCbitless external and opaque AS LSA 0
```

```
Number of DoNotAge external and opaque AS LSA 0
```

```
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
```

```
Number of areas transit capable is 0
```

```
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area 1
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 2d21h ago
SPF algorithm executed 3 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04E6D6
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0
```

```
Switch1(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi1/0/1 6 1 192.168.1.1/30 1 DR 1/1
```

```
Switch1(config)#do show ip ospf neighbor detail
Neighbor 1.1.1.1, interface address 192.168.1.2, interface-id 6
In the area 1 via interface GigabitEthernet1/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.1.1 BDR is 192.168.1.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:31
Neighbor is up for 2d21h
Index 1/1/1, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
```

```
Switch1(config)#do show ip int brief
Interface IP-Address OK? Method Status Protocol
Vlan1 unassigned YES unset administratively down down
GigabitEthernet0/0 unassigned YES unset administratively down
down
GigabitEthernet1/0/1 192.168.1.1 YES manual up up
GigabitEthernet1/0/2 unassigned YES unset down down
GigabitEthernet1/0/3 unassigned YES unset down down
GigabitEthernet1/0/4 unassigned YES unset down down
GigabitEthernet1/0/5 unassigned YES unset down down
```

```
GigabitEthernet1/0/6 unassigned YES unset down down
GigabitEthernet1/0/7 unassigned YES unset down down
GigabitEthernet1/0/8 unassigned YES unset down down
GigabitEthernet1/0/9 unassigned YES unset down down
GigabitEthernet1/0/10 unassigned YES unset down down
GigabitEthernet1/0/11 unassigned YES unset down down
GigabitEthernet1/0/12 unassigned YES unset down down
GigabitEthernet1/0/13 unassigned YES unset down down
GigabitEthernet1/0/14 unassigned YES unset down down
GigabitEthernet1/0/15 unassigned YES unset down down
GigabitEthernet1/0/16 unassigned YES unset down down
GigabitEthernet1/0/17 unassigned YES unset down down
GigabitEthernet1/0/18 unassigned YES unset down down
GigabitEthernet1/0/19 unassigned YES unset down down
GigabitEthernet1/0/20 unassigned YES unset down down
GigabitEthernet1/0/21 unassigned YES unset down down
GigabitEthernet1/0/22 unassigned YES unset down down
GigabitEthernet1/0/23 unassigned YES unset down down
GigabitEthernet1/0/24 unassigned YES unset down down
GigabitEthernet1/1/1 unassigned YES unset down down
GigabitEthernet1/1/2 unassigned YES unset down down
GigabitEthernet1/1/3 unassigned YES unset down down
GigabitEthernet1/1/4 unassigned YES unset down down
```

Switch1(config)#do show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

H - NHRP, G - NHRP registered, g - NHRP registration summary

o - ODR, P - periodic downloaded static route, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from PFR

& - replicated local route overrides by connected

Gateway of last resort is not set

```
192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C   192.168.1.0/30 is directly connected, GigabitEthernet1/0/1
L   192.168.1.1/32 is directly connected, GigabitEthernet1/0/1
O   192.168.1.4/30
    [110/2] via 192.168.1.2, 00:05:12, GigabitEthernet1/0/1
192.168.2.0/30 is subnetted, 2 subnets
O IA 192.168.2.0 [110/3] via 192.168.1.2, 00:05:12,
GigabitEthernet1/0/1
O IA 192.168.2.4 [110/4] via 192.168.1.2, 00:05:12,
GigabitEthernet1/0/1
192.168.3.0/30 is subnetted, 2 subnets
O IA 192.168.3.0 [110/6] via 192.168.1.2, 00:05:12,
GigabitEthernet1/0/1
O IA 192.168.3.4 [110/5] via 192.168.1.2, 00:05:12,
GigabitEthernet1/0/1
```

```
Switch1(config)#do show run
Building configuration...
Current configuration : 9242 bytes
```

Last configuration change at 14:20:20 UTC Mon Sep 15 2025

```
version 17.6
service timestamps debug datetime msec
service timestamps log datetime msec
Call-home is enabled by Smart-Licensing.
service call-home
platform punt-keepalive disable-kernel-core
hostname SWITCH1
vrf definition Mgmt-vrf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
switch 1 provision c9200l-24t-4g
ip routing
login on-success log
ipv6 unicast-routing
crypto pki trustpoint TP-self-signed-1225728909
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1225728909
revocation-check none
rsa-keypair TP-self-signed-1225728909
```

```
crypto pki trustpoint SLA-TrustPoint
enrollment pkcs12
revocation-check crl
crypto pki certificate chain TP-self-signed-1225728909
certificate self-signed 01
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 31323235 37323839 3039301E 170D3235 30393132
31373134
35385A17 0D333530 39313231 37313435 385A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31
32323537
32383930 39308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 0100CF8D 38F49911 E78379AA 83BB6D2E 6330B072 6E563AE7
EE8B257F
B23F6F0A 65D1B398 12E7EA52 078D3F8C 328EA977 FD3AFA88 35D1B67F
B59C6475
905F987C F491B6BB DA294095 A736477E 03AD81F0 71A16BE0 DC49808D
3C21E80A
40AE4E3D 6B04575B 2EDA6A3B 29F26402 4FB58127 4515C5CA 4F4DDB76
2C1D03D1
A1A9BE67 95C59DD2 D4C2DC96 9EA9E6B3 BDDE81CD A5E85EA6 34CF7E1D
857B1DC0
FFD8C190 03DFADB1 4362DD38 F01B11EE EA6CE877 DE46F38F 0C3B784E
C4869413
66BD828A 2880DF4D 597B7375 E02E4D4D 6678CC4B 4C89C3D7 95668399
7192BCD3
4E2E579D E4425C23 3D7A85E4 02AFC8F5 5630E3B5 7C28C562 5CCF123F
DF0FA98B
90583061 B8FD0203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 1476ED6B 8C6D563F A401993B 5807F9C0
FF7F2A6D
17301D06 03551D0E 04160414 76ED6B8C 6D563FA4 01993B58 07F9C0FF
7F2A6D17
300D0609 2A864886 F70D0101 05050003 82010100 60EB3437 55CAAE90
052AE9E3
7CE968F4 69874583 904D0771 42653C46 3000D8D0 52178233 5E37C7D6
6B8B7931
6CF12405 2FE0BCFC B94DCE8F 157BDD85 058032FD 955E330B 9AAC23BE
17BB584B
FA1B38AA 4B5259C1 3FA935D1 44D9F78D 4501D7F5 E9DE33FA C91726FE
56F5272D
```

FEB243B9 025B4165 4CCF1E30 3BF6909C 16887FA2 CB00C472 1A5138F2
1C450CD2
6D9A624B 6D66ECF4 2D03F76F 8426AF2A 1BC0E82C EC59BC11 A582E3A5
30534BC9
96B52780 F1AA4CC6 39C58406 9A4619D3 7AFC8DD9 F2F440B3 4844FC7E
AE8456AF
2529F244 DEBAAF70 5773E702 FA06C0E5 8D6739E3 4EF20215 3196360D
320C15C5
DE96DF55 063AAEF4 6AAF6226 5B601309 4C84C1EF
quit
crypto pki certificate chain SLA-TrustPoint
certificate ca 01
30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101
0B050030
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317
43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533
30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355
040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73
696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382
010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1
F1EFF64D
CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388
8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7
D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F
EA2956AC
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF
58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B
42C68BB7
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8
8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368
95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04
04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604
1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01
010B0500

```
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78
240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1
6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146
8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2
55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49
1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69
39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD
230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F
719BB2F0
D697DF7F 28
quit
license boot level network-essentials addon dna-essentials
diagnostic bootup level minimal
spanning-tree mode rapid-pvst
spanning-tree extend system-id
memory free low-watermark processor 10633
redundancy
mode sso
transceiver type all
monitoring
class-map match-any system-cpp-police-ewlc-control
description EWLC Control
class-map match-any system-cpp-police-topology-control
description Topology control
class-map match-any system-cpp-police-sw-forward
description Sw forwarding, L2 LVX data packets, LOGGING, Transit
Traffic
class-map match-any system-cpp-default
description EWLC data, Inter FED Traffic
class-map match-any system-cpp-police-sys-data
description Openflow, Exception, EGR Exception, NFL Sampled
Data, RPF Failed
class-map match-any system-cpp-police-punt-webauth
description Punt Webauth
class-map match-any system-cpp-police-l2lvx-control
description L2 LVX control packets
class-map match-any system-cpp-police-forus
description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
description MCAST END STATION
```

```
class-map match-any system-cpp-police-high-rate-app
description High Rate Applications
class-map match-any system-cpp-police-multicast
description MCAST Data
class-map match-any system-cpp-police-l2-control
description L2 control
class-map match-any system-cpp-police-dot1x-auth
description DOT1X Auth
class-map match-any system-cpp-police-data
description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
description Stackwise Virtual OOB
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
description DHCP snooping
class-map match-any system-cpp-police-ios-routing
description L2 control, Topology control, Routing control, Low
Latency
class-map match-any system-cpp-police-system-critical
description System Critical and Gold Pkt
class-map match-any system-cpp-police-ios-feature
description
ICMPGEN, BROADCAST, ICMP, L2LVXCntrl, ProtoSnoop, PuntWebauth, MCASTDa
ta, Transit, DOT1XAuth, Swfwd, LOGGING, L2LVXData, ForusTraffic, ForusA
RP, McastEndStn, Openflow, Exception, EGRExcption, NflSampled, RpfFail
ed
policy-map system-cpp-policy
interface GigabitEthernet0/0
vrf forwarding Mgmt-vrf
no ip address
shutdown
interface GigabitEthernet1/0/1
no switchport
ip address 192.168.1.1 255.255.255.252
ip ospf 6 area 1
ipv6 address 2001:DB8:ACAD:1::1/126
ipv6 ospf 6 area 1
interface GigabitEthernet1/0/2
interface GigabitEthernet1/0/3
interface GigabitEthernet1/0/4
interface GigabitEthernet1/0/5
interface GigabitEthernet1/0/6
interface GigabitEthernet1/0/7
```

```
interface GigabitEthernet1/0/8
interface GigabitEthernet1/0/9
interface GigabitEthernet1/0/10
interface GigabitEthernet1/0/11
interface GigabitEthernet1/0/12
interface GigabitEthernet1/0/13
interface GigabitEthernet1/0/14
interface GigabitEthernet1/0/15
interface GigabitEthernet1/0/16
interface GigabitEthernet1/0/17
interface GigabitEthernet1/0/18
interface GigabitEthernet1/0/19
interface GigabitEthernet1/0/20
interface GigabitEthernet1/0/21
interface GigabitEthernet1/0/22
interface GigabitEthernet1/0/23
interface GigabitEthernet1/0/24
interface GigabitEthernet1/1/1
interface GigabitEthernet1/1/2
interface GigabitEthernet1/1/3
interface GigabitEthernet1/1/4
interface Vlan1
no ip address
shutdown
router ospfv3 6
router-id 6.6.6.6
address-family ipv6 unicast
exit-address-family
router ospf 6
router-id 6.6.6.6
network 192.168.1.0 0.0.0.3 area 1
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
control-plane
service-policy input system-cpp-policy
line con 0
stopbits 1
line aux 0
line vty 0 4
login
transport input ssh
line vty 5 15
login
transport input ssh
call-home
```

```
If contact email address in call-home is configured as sch-  
smart-licensing@cisco.com  
the email address configured in Cisco Smart License Portal will  
be used as contact email address to send SCH notifications.  
contact-email-addr sch-smart-licensing@cisco.com  
profile "CiscoTAC-1"  
active  
destination transport-method http  
end
```

Router1:

```
Router1(config)#do show ip ospf  
Routing Process "ospf 1" with ID 1.1.1.1  
Start time: 00:51:58.440, Time elapsed: 4d00h  
Supports only single TOS(TOS0) routes  
Supports opaque LSA  
Supports Link-local Signaling (LLS)  
Supports area transit capability  
Supports NSSA (compatible with RFC 3101)  
Supports Database Exchange Summary List Optimization (RFC 5243)  
Event-log enabled, Maximum number of events: 1000, Mode: cyclic  
Router is not originating router-LSAs with maximum metric  
Initial SPF schedule delay 50 msec  
Minimum hold time between two consecutive SPF 200 msec  
Maximum wait time between two consecutive SPF 5000 msec  
Incremental-SPF disabled  
Initial LSA throttle delay 50 msec  
Minimum hold time for LSA throttle 200 msec  
Maximum wait time for LSA throttle 5000 msec  
Minimum LSA arrival 100 msec  
LSA group pacing timer 240 secs  
Interface flood pacing timer 33 msec  
Retransmission pacing timer 66 msec  
EXCHANGE/LOADING adjacency limit: initial 300, process maximum  
300  
Number of external LSA 0. Checksum Sum 0x000000  
Number of opaque AS LSA 0. Checksum Sum 0x000000  
Number of DCbitless external and opaque AS LSA 0  
Number of DoNotAge external and opaque AS LSA 0  
Number of areas in this router is 1. 1 normal 0 stub 0 nssa  
Number of areas transit capable is 0  
External flood list length 0  
IETF NSF helper support enabled  
Cisco NSF helper support enabled  
Reference bandwidth unit is 100 mbps  
Area 1  
Number of interfaces in this area is 2
```

Area has no authentication
SPF algorithm last executed 2d22h ago
SPF algorithm executed 58 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04E2D8
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Router1(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/1 1 1 192.168.1.5/30 1 DR 1/1
Gi0/0/0 1 1 192.168.1.2/30 1 BDR 1/1

Router1(config)#do show ip ospf neighbor detail
Neighbor 2.2.2.2, interface address 192.168.1.6, interface-id 6
In the area 1 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.1.5 BDR is 192.168.1.6
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:37
Neighbor is up for 4d00h
Index 2/2/2, retransmission queue length 0, number of
retransmission 4
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 2
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 192.168.1.1, interface address 192.168.1.1, interface-
id 38
In the area 1 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.1.1 BDR is 192.168.1.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:38
Neighbor is up for 2d22h
Index 1/1/1, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

```
Router1(config)#do show ip interface brief
Any interface listed with OK? value "NO" does not have a valid
configuration
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.1.2 YES manual up up
GigabitEthernet0/0/1 192.168.1.5 YES manual up up
Serial0/1/0 unassigned NO unset down down
Serial0/1/1 unassigned NO unset down down
GigabitEthernet0/2/0 unassigned YES unset administratively down
down
GigabitEthernet0/2/1 unassigned YES unset administratively down
down
GigabitEthernet0 unassigned YES unset administratively down down
```

```
Router1(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
      D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
      N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
      E1 - OSPF external type 1, E2 - OSPF external type 2
      i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
      ia - IS-IS inter area, * - candidate default, U - per-user
static route
      o - ODR, P - periodic downloaded static route, H - NHRP, l -
LISP
      a - application route
      + - replicated route, % - next hop override, p - overrides
from PfR
```

Gateway of last resort is not set

```
      192.168.1.0/24 is variably subnetted, 4 subnets, 2 masks
C    192.168.1.0/30 is directly connected, GigabitEthernet0/0/0
L    192.168.1.2/32 is directly connected, GigabitEthernet0/0/0
C    192.168.1.4/30 is directly connected, GigabitEthernet0/0/1
L    192.168.1.5/32 is directly connected, GigabitEthernet0/0/1
      192.168.2.0/30 is subnetted, 2 subnets
O IA 192.168.2.0 [110/2] via 192.168.1.6, 3d02h,
GigabitEthernet0/0/1
O IA 192.168.2.4 [110/3] via 192.168.1.6, 2d21h,
GigabitEthernet0/0/1
```

```
192.168.3.0/30 is subnetted, 2 subnets
0 IA 192.168.3.0 [110/5] via 192.168.1.6, 02:04:56,
GigabitEthernet0/0/1
0 IA 192.168.3.4 [110/4] via 192.168.1.6, 02:05:36,
GigabitEthernet0/0/1
```

```
Router1(config)#do show run
Building configuration...
Current configuration : 3990 bytes
Last configuration change at 15:49:23 UTC Thu Sep 11 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router1
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-48848711
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-48848711
revocation-check none
rsakeypair TP-self-signed-48848711
crypto pki certificate chain TP-self-signed-48848711
certificate self-signed 01
3082032C 30820214 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
2F312D30 2B060355 04031324 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 34383834 38373131 301E170D 32353039 31313135
34393234
5A170D33 30303130 31303030 3030305A 302F312D 302B0603 55040313
24494F53
2D53656C 662D5369 676E6564 2D436572 74696669 63617465 2D343838
34383731
```

31308201 22300D06 092A8648 86F70D01 01010500 0382010F 00308201
0A028201
01009F2C E8602FF3 1BB9D7BD 0DA227EE E53AD8C4 19534485 C26B93F8
A5392DCF
A5EB64A8 93E9221B AAC5B478 9F99A92F 88448817 045833A2 0E8782F8
509B4146
BA6D420A C82B01F9 C0571795 2A8D5D6B 6F6C2A88 30DA4D73 7BA2DDD8
266D1870
C311C58B 18E2EA65 1CA4460E 36DC46DB AFA5C343 104F3851 A0D325F1
4C76A576
50A1FD5B 73C5D529 EB53C843 31C2F40A 9D1D51A8 D62BA7D4 18FD4F2D
E7499F8A
F9CF2CE9 B2DA7F7D 3DADE7A2 8EDE63CF 7282F31C BD2FE1C8 F7D8D4F3
2D842DFE
4608A8DC F7DBC80 18C0A205 378A2E5F FD7FB689 642CA2C3 33EA9DAB
43656FD9
9BBF5468 36F1B657 9E3FC51F 91D8BA2A 3BC0602C 25008C08 28BD78AA
5A224834
6BBD0203 010001A3 53305130 0F060355 1D130101 FF040530 030101FF
301F0603
551D2304 18301680 14C7150A D125EB9A 72A307CF 33924973 E073A353
7D301D06
03551D0E 04160414 C7150AD1 25EB9A72 A307CF33 924973E0 73A3537D
300D0609
2A864886 F70D0101 05050003 82010100 1AA8F97C C3FDBF36 588D1D65
EC686BAA
4FAD446C 0086662D 2ABB9EE4 3DBD3D97 5D5AB692 AB5E1DEC D110E877
086879EA
9F6A81EA BC1EA3CA 820E3EED 4BB8E0DE 13671C55 496A8114 4A5A6BA8
1894068D
EBE7D495 F6A00EF4 96B23654 99213BAF 76939456 FA06C1FF 89569438
9012BA6D
9DD03025 20CF31C0 9A689E77 7FD05C6C FB680FB4 8529375E 9CB211C9
8C91F9DB
02DB8066 49DD9971 73823910 FFC8C05C 3608150E 2025B8B8 9D154B78
5B1709D7
FF7599EA 0B9C4B2C E2F039B0 F5A2DA97 17CF0569 E2E574EE DEDA63C8
C6FC18BD
4B102CF5 4FDF2E00 D4E500A1 764B8C73 57BEA0E4 7E22DB39 D6851C34
39923117
84B66321 F399D63C F50ED488 CDD65D3A
quit
license udi pid ISR4321/K9 sn FD0214328HZ
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy

```
mode none
interface GigabitEthernet0/0/0
ip address 192.168.1.2 255.255.255.252
ip ospf 1 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:1::2/126
ipv6 ospf 1 area 1
interface GigabitEthernet0/0/1
ip address 192.168.1.5 255.255.255.252
ip ospf 1 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:1::5/126
ipv6 ospf 1 area 1
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 1
router-id 1.1.1.1
network 192.168.1.0 0.0.0.3 area 1
network 192.168.1.4 0.0.0.3 area 1
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 1
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router2:

```
Router2(config)#do show ip ospf
Routing Process "ospf 2" with ID 2.2.2.2
Start time: 1w1d, Time elapsed: 4d00h
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
It is an area border router
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msec
Minimum hold time between two consecutive SPF 200 msec
Maximum wait time between two consecutive SPF 5000 msec
Incremental-SPF disabled
Initial LSA throttle delay 50 msec
Minimum hold time for LSA throttle 200 msec
Maximum wait time for LSA throttle 5000 msec
Minimum LSA arrival 100 msec
LSA group pacing timer 240 sec
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 2. 2 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area BACKBONE(0)
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 01:31:18.433 ago
SPF algorithm executed 55 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04C5D1
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
```

Number of DoNotAge LSA 0
Flood list length 0
Area 1
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 2d22h ago
SPF algorithm executed 42 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04DEDA
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Router2(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/1 2 0 192.168.2.1/30 1 BDR 1/1
Gi0/0/0 2 1 192.168.1.6/30 1 BDR 1/1

Router2(config)#do show ip ospf neighbor detail
Neighbor 3.3.3.3, interface address 192.168.2.2, interface-id 6
In the area 0 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.2.2 BDR is 192.168.2.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:35
Neighbor is up for 3d01h
Index 1/1/2, retransmission queue length 0, number of
retransmission 4
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 1.1.1.1, interface address 192.168.1.5, interface-id 7
In the area 1 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.1.5 BDR is 192.168.1.6
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:38
Neighbor is up for 4d00h
Index 1/1/1, retransmission queue length 0, number of
retransmission 3
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)

Last retransmission scan length is 2, maximum is 2
Last retransmission scan time is 0 msec, maximum is 0 msec

```
Router2(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.1.6 YES manual up up
GigabitEthernet0/0/1 192.168.2.1 YES manual up up
Serial0/1/0 unassigned YES unset administratively down down
Serial0/1/1 unassigned YES unset administratively down down
GigabitEthernet0 unassigned YES unset administratively down down
```

```
Router2(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
        ia - IS-IS inter area, * - candidate default, U - per-user
static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l -
LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides
from PfR
```

Gateway of last resort is not set

```
192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
O 192.168.1.0/30
   [110/2] via 192.168.1.5, 00:10:52, GigabitEthernet0/0/0
C 192.168.1.4/30 is directly connected, GigabitEthernet0/0/0
L 192.168.1.6/32 is directly connected, GigabitEthernet0/0/0
192.168.2.0/24 is variably subnetted, 3 subnets, 2 masks
C 192.168.2.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.2.1/32 is directly connected, GigabitEthernet0/0/1
O 192.168.2.4/30 [110/2] via 192.168.2.2, 2d21h,
GigabitEthernet0/0/1
192.168.3.0/30 is subnetted, 2 subnets
O IA 192.168.3.0 [110/4] via 192.168.2.2, 02:06:03,
GigabitEthernet0/0/1
```

0 IA 192.168.3.4 [110/3] via 192.168.2.2, 02:06:43,
GigabitEthernet0/0/1

```
Router2(config)#do show run
Building configuration...
Current configuration : 3905 bytes
Last configuration change at 17:01:42 UTC Mon Sep 15 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router2
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
vtp domain cisco
vtp mode transparent
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-1411542050
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1411542050
revocation-check none
rsakeypair TP-self-signed-1411542050
crypto pki certificate chain TP-self-signed-1411542050
certificate self-signed 01
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 31343131 35343230 3530301E 170D3235 30393131
31343233
30395A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31
34313135
34323035 30308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
```

0A028201 01009ADC EF88D7AE 1D3307A2 A5AD1B63 2EF8AAF1 533E9A45
009FD5B8
13D7C12F 7A20141D 8B75E6D3 081A8868 7C753E05 28D34C83 51929A33
8DB44D19
14E892B1 34A41170 E499A5CE E804F3CE 4FC26B76 A8CA4C00 48FE969A
2CD0DD93
68FCA989 DE8EAC7C 7EEEC39C EAEC36D5 F3419771 4C7EBBE1 6CAD6483
A610032A
F153A90F 663A9CC0 2CD07B9F 53391363 62E16AC4 9F30BFD6 F24B1DFA
2ED0523D
9D74727A 91B3CE02 CA8B2E1F 80855719 B7EFC54F 6417F72C 2B222A2D
F756AD38
44239C20 93084A42 6FF78C9C D99F84EB D9012FD5 529956A8 839E6C95
AF5A3B0E
3F6A67F7 0E3B4A88 401C898B 15B85941 93248370 3A76B838 9D72B1EF
6BA882FB
297A74D6 B9690203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 14C82A68 7DE5B186 FE679B1C 30F9A7F6
64B3635A
F0301D06 03551D0E 04160414 C82A687D E5B186FE 679B1C30 F9A7F664
B3635AF0
300D0609 2A864886 F70D0101 05050003 82010100 8647BDFC FE685CE2
E3686C2A
CD052A1B A0D2356C 294639A8 6E0B3627 2A7AFF6C 1BE145A5 6AB049DC
43A326CC
475AA583 C5CEA83B 6066994F A7BF408B 5AA6499E 3C9119C8 AE345BEC
9031E689
8DFDA2B1 5F5A30FE 844BF07A 0082D6F3 2B85B989 13774E60 BFC7325F
B95A5B95
CEA03292 8E5CF852 D4CA7ACE B4239E1E CE5C273A F64431C9 2767039A
912EEC15
3E398B74 450FB172 382248B1 1DB6CA89 BE26B357 B9ADC191 C7E8E723
E0173092
3CA9E875 FF470610 CB5F2DF2 4732D227 F3DB0A8D 59162E4E 675F54D3
F303770A
B1E4296D 52E04E92 F9FE0E8B 1CE188EF 114740FB 651EB1E2 B45051BD
2A58A3C2
45A36638 DDB29AAE 13915261 58DE0E1A 6823749C

quit

license udi pid ISR4321/K9 sn FD0215009PN

no license smart enable

diagnostic bootup level minimal

spanning-tree extend system-id

redundancy

mode none

interface GigabitEthernet0/0/0

```
ip address 192.168.1.6 255.255.255.252
ip ospf 2 area 1
negotiation auto
ipv6 address 2001:DB8:ACAD:1::6/126
ipv6 ospf 2 area 1
interface GigabitEthernet0/0/1
ip address 192.168.2.1 255.255.255.252
ip ospf 2 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::1/126
ipv6 ospf 2 area 0
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 2
router-id 2.2.2.2
network 192.168.1.4 0.0.0.3 area 1
network 192.168.2.0 0.0.0.3 area 0
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 2
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Area 0:

Router3:

```
Router3(config)#do show ip ospf
Routing Process "ospf 3" with ID 3.3.3.3
Start time: 01:13:39.058, Time elapsed: 4d00h
```

Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msec
Minimum hold time between two consecutive SPF's 200 msec
Maximum wait time between two consecutive SPF's 5000 msec
Incremental-SPF disabled
Initial LSA throttle delay 50 msec
Minimum hold time for LSA throttle 200 msec
Maximum wait time for LSA throttle 5000 msec
Minimum LSA arrival 100 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
EXCHANGE/LOADING adjacency limit: initial 300, process maximum 300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area BACKBONE(0)
Number of interfaces in this area is 2
Area has no authentication
SPF algorithm last executed 01:33:53.510 ago
SPF algorithm executed 56 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04C5D1
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Router3(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/1 3 0 192.168.2.5/30 1 BDR 1/1

```
Gi0/0/0 3 0 192.168.2.2/30 1 DR 1/1
```

```
Router3(config)#do show ip ospf neighbor detail
Neighbor 192.168.2.6, interface address 192.168.2.6, interface-
id 6
In the area 0 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.2.6 BDR is 192.168.2.5
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:38
Neighbor is up for 2d20h
Index 1/2/2, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 2.2.2.2, interface address 192.168.2.1, interface-id 7
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.2.2 BDR is 192.168.2.1
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:32
Neighbor is up for 3d01h
Index 1/1/1, retransmission queue length 0, number of
retransmission 11
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
Last retransmission scan time is 0 msec, maximum is 0 msec
```

```
Router3(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.2.2 YES manual up up
GigabitEthernet0/0/1 192.168.2.5 YES manual up up
Serial0/1/0 unassigned YES unset down down
Serial0/1/1 unassigned YES unset down down
GigabitEthernet0/2/0 unassigned YES unset down down
GigabitEthernet0/2/1 unassigned YES unset down down
GigabitEthernet0 unassigned YES unset down down
```

```
Router3(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
```

area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

192.168.1.0/30 is subnetted, 2 subnets
O IA 192.168.1.0 [110/3] via 192.168.2.1, 00:12:37, GigabitEthernet0/0/0
O IA 192.168.1.4 [110/2] via 192.168.2.1, 3d02h, GigabitEthernet0/0/0
192.168.2.0/24 is variably subnetted, 4 subnets, 2 masks
C 192.168.2.0/30 is directly connected, GigabitEthernet0/0/0
L 192.168.2.2/32 is directly connected, GigabitEthernet0/0/0
C 192.168.2.4/30 is directly connected, GigabitEthernet0/0/1
L 192.168.2.5/32 is directly connected, GigabitEthernet0/0/1
192.168.3.0/30 is subnetted, 2 subnets
O IA 192.168.3.0 [110/3] via 192.168.2.6, 02:07:17, GigabitEthernet0/0/1
O IA 192.168.3.4 [110/2] via 192.168.2.6, 02:07:57, GigabitEthernet0/0/1

Router3(config)#do show run

Building configuration...

Current configuration : 1698 bytes

Last configuration change at 15:26:04 UTC Mon Sep 15 2025

version 16.9

service timestamps debug datetime msec

service timestamps log datetime msec

platform qfp utilization monitor load 80

platform punt-keepalive disable-kernel-core

hostname Router3

boot-start-marker

boot-end-marker

```
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FD0214420HS
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.2.2 255.255.255.252
ip ospf 3 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::2/126
ipv6 ospf 3 area 0
interface GigabitEthernet0/0/1
ip address 192.168.2.5 255.255.255.252
ip ospf 3 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::5/126
ipv6 ospf 3 area 0
interface Serial0/1/0
no ip address
interface Serial0/1/1
no ip address
interface GigabitEthernet0/2/0
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
router ospf 3
router-id 3.3.3.3
network 192.168.2.0 0.0.0.3 area 0
network 192.168.2.4 0.0.0.3 area 0
ip forward-protocol nd
```

```
no ip http server
ip http secure-server
ipv6 router ospf 3
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
End
```

Area 2:

Router4:

```
Router4(config)#do show ip ospf
Routing Process "ospf 4" with ID 192.168.2.6
Start time: 00:28:28.630, Time elapsed: 2d21h
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
It is an area border router
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msec
Minimum hold time between two consecutive SPF 200 msec
Maximum wait time between two consecutive SPF 5000 msec
Incremental-SPF disabled
Initial LSA throttle delay 50 msec
Minimum hold time for LSA throttle 200 msec
Maximum wait time for LSA throttle 5000 msec
Minimum LSA arrival 100 msec
LSA group pacing timer 240 sec
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 2. 2 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
```

```
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area BACKBONE(0)
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 02:04:40.290 ago
SPF algorithm executed 28 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04B1DB
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0
Area 2
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 02:03:44.585 ago
SPF algorithm executed 49 times
Area ranges are
Number of LSA 9. Checksum Sum 0x047BC2
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0
```

```
Router4(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/0 4 0 192.168.2.6/30 1 DR 1/1
Gi0/0/1 4 2 192.168.3.6/30 1 DR 1/1
```

```
Router4(config)#do show ip ospf neighbor detail
Neighbor 3.3.3.3, interface address 192.168.2.5, interface-id 7
In the area 0 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.2.6 BDR is 192.168.2.5
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:37
Neighbor is up for 2d20h
Index 1/1/1, retransmission queue length 0, number of
retransmission 4
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 1, maximum is 1
```

Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 5.5.5.5, interface address 192.168.3.5, interface-id 6
In the area 2 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.3.6 BDR is 192.168.3.5
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:31
Neighbor is up for 01:36:29
Index 1/1/2, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec

Router4(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.2.6 YES manual up up
GigabitEthernet0/0/1 192.168.3.6 YES manual up up
GigabitEthernet0 unassigned YES unset administratively down down

Router4(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
ia - IS-IS inter area, * - candidate default, U - per-user
static route
o - ODR, P - periodic downloaded static route, H - NHRP, l -
LISP
a - application route
+ - replicated route, % - next hop override, p - overrides
from PfR

Gateway of last resort is not set

192.168.1.0/30 is subnetted, 2 subnets
0 IA 192.168.1.0 [110/4] via 192.168.2.5, 00:13:14,
GigabitEthernet0/0/0

```
O IA 192.168.1.4 [110/3] via 192.168.2.5, 02:08:34,
GigabitEthernet0/0/0
    192.168.2.0/24 is variably subnetted, 3 subnets, 2 masks
O    192.168.2.0/30
    [110/2] via 192.168.2.5, 02:08:34, GigabitEthernet0/0/0
C    192.168.2.4/30 is directly connected, GigabitEthernet0/0/0
L    192.168.2.6/32 is directly connected, GigabitEthernet0/0/0
    192.168.3.0/24 is variably subnetted, 3 subnets, 2 masks
O    192.168.3.0/30
    [110/2] via 192.168.3.5, 02:07:48, GigabitEthernet0/0/1
C    192.168.3.4/30 is directly connected, GigabitEthernet0/0/1
L    192.168.3.6/32 is directly connected, GigabitEthernet0/0/1
```

```
Router4(config)#do show run
Building configuration...
Current configuration : 3725 bytes
Last configuration change at 20:15:08 UTC Fri Sep 12 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router4
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-187689846
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-187689846
revocation-check none
rsakeypair TP-self-signed-187689846
crypto pki certificate chain TP-self-signed-187689846
certificate self-signed 01
3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
```

30312E30	2C060355	04031325	494F532D	53656C66	2D536967	6E65642D
43657274						
69666963	6174652D	31383736	38393834	36301E17	0D323530	39313231
39323630						
315A170D	33303031	30313030	30303030	5A303031	2E302C06	03550403
1325494F						
532D5365	6C662D53	69676E65	642D4365	72746966	69636174	652D3138
37363839						
38343630	82012230	0D06092A	864886F7	0D010101	05000382	010F0030
82010A02						
82010100	C4FFA596	31767A3A	04BB0320	8D4CA352	D8D3811B	7590CDA5
BADB81C6						
634CC332	CBC6EE98	8BA62B22	BADA051D	7DB75D82	7E57AE87	87B8CF0A
213A57BC						
92D7DCD4	EE7C63D1	02E15906	546C5858	0F90B11F	8045CE69	F451F547
FA090C61						
B8C85427	D6EA2733	50BE25C8	302651E7	FA297284	3A41AE84	7ACBB6A3
72146FC0						
4AF59E0E	163981F0	96591D27	07B44A37	47617426	6C83BBA9	A782A1B9
9E1B0412						
FB809A24	F2809526	CBF14D46	ADA3EAF7	BCDB2BE3	AA230BC3	F2BCB214
22FE8911						
D24E1461	7AA70D10	6D6A25D8	4DE942EB	4D87C682	AE8D1F31	ED73892A
67B8C9DB						
2B387F3D	A867EA5E	EBAEAC6E	4453AF5A	3AA99910	4F9ACC16	40135F9B
1FA839D3						
C8E0EA7B	02030100	01A35330	51300F06	03551D13	0101FF04	05300301
01FF301F						
0603551D	23041830	168014D5	07205C88	D0577F79	9EDE4B39	EE754EC9
E1AAD130						
1D060355	1D0E0416	0414D507	205C88D0	577F799E	DE4B39EE	754EC9E1
AAD1300D						
06092A86	4886F70D	01010505	00038201	010014E5	1BDDAD2A	A2617015
896ABC0E						
0352CC43	8C628996	D28D2539	2B6C1E4A	D67D032F	D6BBDDF7	1626BA0E
76A17267						
52CEA9EB	907A3005	F5027669	740738FD	3C406ED8	689B7F3A	89B11424
E19C348A						
27551845	DAFB05BF	05667D2C	D7F6E910	8DD54ABF	7B4BD109	1C8E8C51
7BE5188A						
F814018F	1C64B003	61880BC3	E2C30590	0FE35C7B	C47338C1	0791B974
BB788A6C						
D2529CD7	A3755EDA	1DA626E1	D5ECA1F8	720005D2	5043833C	5F403F64
7D7F85B7						
2A915CE2	7F520496	58CB4F38	32E3F134	32B3F13F	F3E400B1	34453C8C
8F97B58F						

```
2424EC2C D292F07C E0F2D94D DB4723BE D178A846 49F74414 9E4FCB0E
4AC30F6D
F447D758 97BBBBD3 CE6852CA 9E812AB4 204E
quit
license udi pid ISR4321/K9 sn FLM2407014H
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.2.6 255.255.255.252
ip ospf 4 area 0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::6/126
ipv6 ospf 4 area 0
interface GigabitEthernet0/0/1
ip address 192.168.3.6 255.255.255.252
ip ospf 4 area 2
negotiation auto
ipv6 address 2001:DB8:ACAD:3::6/126
ipv6 ospf 4 area 2
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 4
router-id 4.4.4.4
network 192.168.2.4 0.0.0.3 area 2
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 4
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router5:

```
Router5(config)#do show ip ospf
Routing Process "ospf 5" with ID 5.5.5.5
Start time: 00:06:12.168, Time elapsed: 02:05:42.612
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msec
Minimum hold time between two consecutive SPF's 200 msec
Maximum wait time between two consecutive SPF's 5000 msec
Incremental-SPF disabled
Initial LSA throttle delay 50 msec
Minimum hold time for LSA throttle 200 msec
Maximum wait time for LSA throttle 5000 msec
Minimum LSA arrival 100 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
EXCHANGE/LOADING adjacency limit: initial 300, process maximum
300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area 2
Number of interfaces in this area is 2
Area has no authentication
SPF algorithm last executed 02:04:47.661 ago
SPF algorithm executed 9 times
Area ranges are
Number of LSA 9. Checksum Sum 0x047BC2
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0
```

```
Router5(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C
Gi0/0/1 5 2 192.168.3.2/30 1 BDR 1/1
Gi0/0/0 5 2 192.168.3.5/30 1 BDR 1/1
```

```
Router5(config)#do show ip ospf neighbor detail
Neighbor 192.168.3.1, interface address 192.168.3.1, interface-
id 38
In the area 2 via interface GigabitEthernet0/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.3.1 BDR is 192.168.3.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:31
Neighbor is up for 02:05:44
Index 1/2/2, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
Neighbor 192.168.2.6, interface address 192.168.3.6, interface-
id 7
In the area 2 via interface GigabitEthernet0/0/0
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.3.6 BDR is 192.168.3.5
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:31
Neighbor is up for 02:05:46
Index 1/1/1, retransmission queue length 0, number of
retransmission 1
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 2, maximum is 2
Last retransmission scan time is 0 msec, maximum is 0 msec
```

```
Router5(config)#do show ip interface brief
Any interface listed with OK? value "NO" does not have a valid
configuration
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.3.5 YES manual up up
GigabitEthernet0/0/1 192.168.3.2 YES manual up up
Serial0/1/0 unassigned NO unset down down
Serial0/1/1 unassigned NO unset down down
GigabitEthernet0 unassigned YES unset administratively down down
```

```
Router5(config)#do show ip route
Codes: L - local, C - connected, S - static, R - RIP, M -
mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
ia - IS-IS inter area, * - candidate default, U - per-user
static route
o - ODR, P - periodic downloaded static route, H - NHRP, l -
LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from
PfR
Gateway of last resort is not set
192.168.1.0/30 is subnetted, 2 subnets
O IA 192.168.1.0 [110/5] via 192.168.3.6, 00:10:50,
GigabitEthernet0/0/0
O IA 192.168.1.4 [110/4] via 192.168.3.6, 02:05:30,
GigabitEthernet0/0/0
192.168.2.0/30 is subnetted, 2 subnets
O IA 192.168.2.0 [110/3] via 192.168.3.6, 02:05:30,
GigabitEthernet0/0/0
O IA 192.168.2.4 [110/2] via 192.168.3.6, 02:05:30,
GigabitEthernet0/0/0
192.168.3.0/24 is variably subnetted, 4 subnets, 2 masks
C 192.168.3.0/30 is directly connected, GigabitEthernet0/0/1
L 192.168.3.2/32 is directly connected, GigabitEthernet0/0/1
C 192.168.3.4/30 is directly connected, GigabitEthernet0/0/0
L 192.168.3.5/32 is directly connected, GigabitEthernet0/0/0
```

```
Router5(config)#do show run
Building configuration...
Current configuration : 3745 bytes
Last configuration change at 17:39:07 UTC Mon Sep 15 2025
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router5
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
```

```
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-1541607431
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1541607431
revocation-check none
rsa-keypair TP-self-signed-1541607431
crypto pki certificate chain TP-self-signed-1541607431
certificate self-signed 01
30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 31353431 36303734 3331301E 170D3235 30393135
31353339
32385A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31
35343136
30373433 31308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
0A028201 01009F52 4BD39D6D 7E4029DD 3EDCA271 EDD29A5C 5CC2EEE0
5AB308BB
1D061F9A 925DDA4A D4565D1D AA00EF25 4877C9D6 4686A9B3 021FC74D
99D86211
56E0E14E CB4FB477 9EB9F9DB C49CBDCD FFE99AD5 C42A831A DBB6660F
6653A69D
21A3504D B141E995 5D3AC8AA BAF4E090 885447E7 CFC3A90E 903CDEAE
63C3782E
2D7088B5 47C6D50C 38C55066 FE9361E5 26B0187A CD4003DA A6315CE2
AE959813
ECF38E93 96FB06AA E7E08F5A A25E4E78 191B35EA 4035F053 C4FD8892
C88BBE9D
8E8707E9 4FD1E4B6 5B7695E9 10A4E80C D7B96B7A AFEC92FC 891A946E
FC78574D
0CC6D5C2 5B1427B3 916CCE7D C1DD9F4F 41E950D4 8B05594A 64312D95
73EF3AF4
505F6DEF D1430203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
301F0603 551D2304 18301680 1464048B 5A670DC1 4F11EE80 759EDFF4
B4ABADB2
```

32301D06 03551D0E 04160414 64048B5A 670DC14F 11EE8075 9EDFF4B4
ABADB232
300D0609 2A864886 F70D0101 05050003 82010100 3455B7E7 926EA442
3C3194FB
9A130641 2F8CB5A5 2D73DC58 73F0D6D4 197ADB96 B7C660A4 1F5A920D
D2FD64D7
F9C1B0E0 BE1A28B7 C7CAF9EF 1FB6BE85 074B3ACB 06423C6D DCF21F66
4F53AABB
570EE013 00964146 0BD77AE3 54CEAAF0 B321973D 52D43B43 8E8EF664
260C190E
93E2DB21 1B644112 2884CA14 08A1BC46 6F768179 89A6014B 8ED4C0D4
30D16E87
E1CD54FD E0DB4CAE 4742815C B3ECD751 FB303A18 D43EC6C7 6228C1AA
B9DF6E37
242226F0 03D16246 32E22BEA DD54D97B 9F108B1C 9AC3F929 C2F302FA
087217FF
FBACABBC E81128D7 1F009FDF 4B538A08 53B3CADB D106D643 E339FDB9
227E5771
1D06A401 648DF9C9 1E52E7CB 9F1F1DE4 7D78146F
quit
license udi pid ISR4321/K9 sn FD0214420GM
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.3.5 255.255.255.252
ip ospf 5 area 2
negotiation auto
ipv6 address 2001:DB8:ACAD:3::5/126
ipv6 ospf 5 area 2
interface GigabitEthernet0/0/1
ip address 192.168.3.2 255.255.255.252
ip ospf 5 area 2
negotiation auto
ipv6 address 2001:DB8:ACAD:3::2/126
ipv6 ospf 5 area 2
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospf 5
router-id 5.5.5.5

```
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router ospf 5
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Switch2:

*Sep 15 16:47:34.882: %SYS-5-CONFIG_I: Configured from console by console

Switch2>

Switch2>en

Switch2#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch2(config)#do traceroute 192.168.1.1

Type escape sequence to abort.

Tracing the route to 192.168.1.1

VRF info: (vrf in name/id, vrf out name/id)

```
 1 192.168.3.2 4 msec 0 msec 4 msec
 2 192.168.3.6 0 msec 4 msec 0 msec
 3 192.168.2.5 4 msec 4 msec 0 msec
 4 192.168.2.1 0 msec 4 msec 0 msec
 5 192.168.1.5 4 msec 4 msec 0 msec
 6 192.168.1.1 4 msec 4 msec *
```

Switch2(config)#do traceroute 2001:db8:acad:1::1

Type escape sequence to abort.

Tracing the route to 2001:DB8:ACAD:1::1

```
 1 2001:DB8:ACAD:3::2 0 msec 0 msec 4 msec
 2 2001:DB8:ACAD:3::6 0 msec 4 msec 0 msec
 3 2001:DB8:ACAD:2::5 0 msec 4 msec 0 msec
 4 2001:DB8:ACAD:2::1 4 msec 4 msec 0 msec
 5 2001:DB8:ACAD:1::5 0 msec 4 msec 0 msec
 6 2001:DB8:ACAD:1::1 4 msec 4 msec 0 msec
```

Switch2(config)#do show ip ospf

Routing Process "ospf 7" with ID 192.168.3.1

Start time: 00:13:22.384, Time elapsed: 2d23h
Supports only single TOS(TOS0) routes
Supports opaque LSA
Supports Link-local Signaling (LLS)
Supports area transit capability
Supports NSSA (compatible with RFC 3101)
Supports Database Exchange Summary List Optimization (RFC 5243)
Event-log enabled, Maximum number of events: 1000, Mode: cyclic
Router is not originating router-LSAs with maximum metric
Initial SPF schedule delay 50 msec
Minimum hold time between two consecutive SPFs 200 msec
Maximum wait time between two consecutive SPFs 5000 msec
Incremental-SPF disabled
Initial LSA throttle delay 50 msec
Minimum hold time for LSA throttle 200 msec
Maximum wait time for LSA throttle 5000 msec
Minimum LSA arrival 100 msec
LSA group pacing timer 240 secs
Interface flood pacing timer 33 msec
Retransmission pacing timer 66 msec
EXCHANGE/LOADING adjacency limit: initial 300, process maximum 300
Number of external LSA 0. Checksum Sum 0x000000
Number of opaque AS LSA 0. Checksum Sum 0x000000
Number of DCbitless external and opaque AS LSA 0
Number of DoNotAge external and opaque AS LSA 0
Number of areas in this router is 1. 1 normal 0 stub 0 nssa
Number of areas transit capable is 0
External flood list length 0
IETF NSF helper support enabled
Cisco NSF helper support enabled
Reference bandwidth unit is 100 mbps
Area 2
Number of interfaces in this area is 1
Area has no authentication
SPF algorithm last executed 01:45:25.520 ago
SPF algorithm executed 50 times
Area ranges are
Number of LSA 9. Checksum Sum 0x04823F
Number of opaque link LSA 0. Checksum Sum 0x000000
Number of DCbitless LSA 0
Number of indication LSA 0
Number of DoNotAge LSA 0
Flood list length 0

Switch2(config)#do show ip ospf interface brief
Interface PID Area IP Address/Mask Cost State Nbrs F/C

```
Gi1/0/1 7 2 192.168.3.1/30 1 DR 1/1
```

```
Switch2(config)#do show ip ospf neighbor detail
Neighbor 5.5.5.5, interface address 192.168.3.2, interface-id 7
In the area 2 via interface GigabitEthernet1/0/1
Neighbor priority is 1, State is FULL, 6 state changes
DR is 192.168.3.1 BDR is 192.168.3.2
Options is 0x12 in Hello (E-bit, L-bit)
Options is 0x52 in DBD (E-bit, L-bit, O-bit)
LLS Options is 0x1 (LR)
Dead timer due in 00:00:38
Neighbor is up for 01:46:24
Index 1/1/1, retransmission queue length 0, number of
retransmission 0
First 0x0(0)/0x0(0)/0x0(0) Next 0x0(0)/0x0(0)/0x0(0)
Last retransmission scan length is 0, maximum is 0
Last retransmission scan time is 0 msec, maximum is 0 msec
```

```
Switch2(config)#do show ip interface brief
Interface IP-Address OK? Method Status Protocol
Vlan1 unassigned YES unset up down
GigabitEthernet0/0 unassigned YES unset administratively down
down
GigabitEthernet1/0/1 192.168.3.1 YES manual up up
GigabitEthernet1/0/2 unassigned YES unset down down
GigabitEthernet1/0/3 unassigned YES unset down down
GigabitEthernet1/0/4 unassigned YES unset down down
GigabitEthernet1/0/5 unassigned YES unset down down
GigabitEthernet1/0/6 unassigned YES unset down down
GigabitEthernet1/0/7 unassigned YES unset down down
GigabitEthernet1/0/8 unassigned YES unset down down
GigabitEthernet1/0/9 unassigned YES unset down down
GigabitEthernet1/0/10 unassigned YES unset down down
GigabitEthernet1/0/11 unassigned YES unset down down
GigabitEthernet1/0/12 unassigned YES unset down down
GigabitEthernet1/0/13 unassigned YES unset down down
GigabitEthernet1/0/14 unassigned YES unset down down
GigabitEthernet1/0/15 unassigned YES unset down down
GigabitEthernet1/0/16 unassigned YES unset down down
GigabitEthernet1/0/17 unassigned YES unset down down
GigabitEthernet1/0/18 unassigned YES unset down down
GigabitEthernet1/0/19 unassigned YES unset down down
GigabitEthernet1/0/20 unassigned YES unset down down
GigabitEthernet1/0/21 unassigned YES unset down down
GigabitEthernet1/0/22 unassigned YES unset down down
GigabitEthernet1/0/23 unassigned YES unset down down
GigabitEthernet1/0/24 unassigned YES unset down down
```

```
GigabitEthernet1/1/1 unassigned YES unset down down
GigabitEthernet1/1/2 unassigned YES unset down down
GigabitEthernet1/1/3 unassigned YES unset down down
GigabitEthernet1/1/4 unassigned YES unset down down
```

```
Switch2(config)#do show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M -  
mobile, B - BGP
```

```
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter  
area
```

```
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external  
type 2
```

```
        E1 - OSPF external type 1, E2 - OSPF external type 2, m -  
OMP
```

```
        n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 -  
IS-IS level-2
```

```
        ia - IS-IS inter area, * - candidate default, U - per-  
user static route
```

```
        H - NHRP, G - NHRP registered, g - NHRP registration  
summary
```

```
        o - ODR, P - periodic downloaded static route, l - LISP  
a - application route
```

```
        + - replicated route, % - next hop override, p -  
overrides from PFR
```

```
        & - replicated local route overrides by connected
```

```
Gateway of last resort is not set
```

```
        192.168.1.0/30 is subnetted, 2 subnets  
O IA      192.168.1.0 [110/6] via 192.168.3.2, 00:31:03,  
GigabitEthernet1/0/1
```

```
O IA      192.168.1.4 [110/5] via 192.168.3.2, 02:25:42,  
GigabitEthernet1/0/1
```

```
        192.168.2.0/30 is subnetted, 2 subnets  
O IA      192.168.2.0 [110/4] via 192.168.3.2, 02:25:42,  
GigabitEthernet1/0/1
```

```
O IA      192.168.2.4 [110/3] via 192.168.3.2, 02:25:42,  
GigabitEthernet1/0/1
```

```
        192.168.3.0/24 is variably subnetted, 3 subnets, 2 masks  
C          192.168.3.0/30 is directly connected,  
GigabitEthernet1/0/1
```

```
L          192.168.3.1/32 is directly connected,  
GigabitEthernet1/0/1
```

```
O          192.168.3.4/30  
            [110/2] via 192.168.3.2, 02:25:42,  
GigabitEthernet1/0/1
```

```
Switch2(config)#do show run
Building configuration...
Current configuration : 9182 bytes
Last configuration change at 14:34:53 UTC Mon Sep 15 2025
version 17.3
service timestamps debug datetime msec
service timestamps log datetime msec
Call-home is enabled by Smart-Licensing.
service call-home
platform punt-keepalive disable-kernel-core
hostname SWITCH2
vrf definition Mgmt-vrf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
switch 1 provision c9200l-24t-4g
ip routing
login on-success log
ipv6 unicast-routing
no device-tracking logging theft
crypto pki trustpoint SLA-TrustPoint
enrollment pkcs12
revocation-check crl
crypto pki trustpoint TP-self-signed-733660727
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-733660727
revocation-check none
rsa-keypair TP-self-signed-733660727
crypto pki certificate chain SLA-TrustPoint
certificate ca 01
30820321 30820209 A0030201 02020101 300D0609 2A864886 F70D0101
0B050030
32310E30 0C060355 040A1305 43697363 6F312030 1E060355 04031317
43697363
6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31 33303533
30313934
3834375A 170D3338 30353330 31393438 34375A30 32310E30 0C060355
040A1305
43697363 6F312030 1E060355 04031317 43697363 6F204C69 63656E73
696E6720
526F6F74 20434130 82012230 0D06092A 864886F7 0D010101 05000382
010F0030
82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 17222EA1
F1EFF64D
```

CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 9CAE6388
8A38E520
1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 700A8BF7
D8F256EE
4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 104FDC5F
EA2956AC
7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 C0BD23CF
58BD7188
68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 539BA42B
42C68BB7
C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 5D5D5FB8
8F27D191
C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B DF5F4368
95135E44
DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 0101FF04
04030201
06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 0E041604
1449DC85
4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 86F70D01
010B0500
03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 49631C78
240DA905
604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 9093D3B1
6C9E3D8B
D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 5575B146
8DFC66A8
467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 11BA9CD2
55A9232C
7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 C71E3B49
1765308B
5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 C37C1E69
39F08678
80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A B623CDBD
230E3AFB
418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 0275156F
719BB2F0
D697DF7F 28
quit
crypto pki certificate chain TP-self-signed-733660727
certificate self-signed 01
3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D
43657274
69666963 6174652D 37333336 36303732 37301E17 0D323530 39313231
36313731

375A170D 33353039 31323136 31373137 5A303031 2E302C06 03550403
1325494F
532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3733
33363630
37323730 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
82010A02
82010100 B854E376 16159541 0DC4254C 264A0048 EEF33ED8 D86CD3BF
0E6E7318
F9CAA299 1F6A70BE 0DC72B46 C874C937 CDB8B682 2578E815 3099FC7D
B0878F40
BD518162 09EF524C 33C54939 E66B93CB 4336C9F4 A210982D 02E66A18
BF0BAB77
EF02DC92 C95B6AD7 7E21F392 9A7C9B8C 2D9F5A15 10A897A9 49B4A52C
806698BB
6948C1E4 9E9C2366 593F392B 5AAFB9BD 98F54F2C EDBC5CFC 0FC58C99
4B82E966
AB72FB0C A1ACE3C4 2841A419 0C293403 578D2315 6AE2E950 C9394D83
36D7548F
95F44C6B D34526C0 2FD1F328 492E6E1D EC3FE764 292AD48B 46323DDD
E8E5B65A
443B8607 65AB8D51 AA12FFB9 4A54ACF2 9A832775 DE7E514C 1B05E286
A432D1D8
1A918023 02030100 01A35330 51300F06 03551D13 0101FF04 05300301
01FF301F
0603551D 23041830 16801471 E8B29DA9 F4C6FCA9 53AB647A 4D899C70
F6C9CC30
1D060355 1D0E0416 041471E8 B29DA9F4 C6FCA953 AB647A4D 899C70F6
C9CC300D
06092A86 4886F70D 01010505 00038201 01002CCC DEDA8DB9 E5EE7F34
09B33AC4
3B69CB23 B187D111 AEADBF80 13502C0E 477FC370 E76587C1 9BC5109E
DFEA589F
32C1A677 0CFBB2E0 F6414D62 092E1D9B 66E73478 167726F9 097AA619
946F9B23
D323FF99 E60165FE 7CCC3F8D F9A1AE93 13DBA2CF E97FB766 C0DBA947
A0576A61
425E3392 E8E819D9 9C8D5409 20FCF85A AD8D6720 ADACE366 D9BCA837
C7FE743D
B9048EFF 8F5DCBBA A3131540 317FD540 EF83454F B715AEE0 F4B49C60
0F47E7A0
511458B6 35DF99EA A772D75F 25B22AFC 5B8C848E AF08910A 53BB2333
8753A36A
34E83F10 2BE30F8A B17EBC8D 6FF36010 0D581945 DF5E68A7 044DD9E7
C85CB893
53848ACF FDFD226A AC83C7D0 4584EDCA B852
quit
license boot level network-essentials addon dna-essentials

```
diagnostic bootup level minimal
spanning-tree mode rapid-pvst
spanning-tree extend system-id
memory free low-watermark processor 10308
redundancy
mode sso
transceiver type all
monitoring
class-map match-any system-cpp-police-ewlc-control
description EWLC Control
class-map match-any system-cpp-police-topology-control
description Topology control
class-map match-any system-cpp-police-sw-forward
description Sw forwarding, L2 LVX data packets, LOGGING, Transit
Traffic
class-map match-any system-cpp-default
description EWLC data, Inter FED Traffic
class-map match-any system-cpp-police-sys-data
description Openflow, Exception, EGR Exception, NFL Sampled
Data, RPF Failed
class-map match-any system-cpp-police-punt-webauth
description Punt Webauth
class-map match-any system-cpp-police-l2lvx-control
description L2 LVX control packets
class-map match-any system-cpp-police-forus
description Forus Address resolution and Forus traffic
class-map match-any system-cpp-police-multicast-end-station
description MCAST END STATION
class-map match-any system-cpp-police-high-rate-app
description High Rate Applications
class-map match-any system-cpp-police-multicast
description MCAST Data
class-map match-any system-cpp-police-l2-control
description L2 control
class-map match-any system-cpp-police-dot1x-auth
description DOT1X Auth
class-map match-any system-cpp-police-data
description ICMP redirect, ICMP_GEN and BROADCAST
class-map match-any system-cpp-police-stackwise-virt-control
description Stackwise Virtual OOB
class-map match-any non-client-nrt-class
class-map match-any system-cpp-police-routing-control
description Routing control and Low Latency
class-map match-any system-cpp-police-protocol-snooping
description Protocol snooping
class-map match-any system-cpp-police-dhcp-snooping
description DHCP snooping
```

```
class-map match-any system-cpp-police-ios-routing
description L2 control, Topology control, Routing control, Low
Latency
class-map match-any system-cpp-police-system-critical
description System Critical and Gold Pkt
class-map match-any system-cpp-police-ios-feature
description
ICMPGEN, BROADCAST, ICMP, L2LVXCntrl, ProtoSnoop, PuntWebauth, MCASTDa
ta, Transit, DOT1XAuth, Swfwd, LOGGING, L2LVXData, ForusTraffic, ForusA
RP, McastEndStn, Openflow, Exception, EGRExcption, NflSampled, RpfFail
ed
policy-map system-cpp-policy
interface GigabitEthernet0/0
vrf forwarding Mgmt-vrf
no ip address
shutdown
negotiation auto
interface GigabitEthernet1/0/1
no switchport
ip address 192.168.3.1 255.255.255.252
ip ospf 7 area 2
ipv6 address 2001:DB8:ACAD:3::1/126
ipv6 ospf 7 area 2
interface GigabitEthernet1/0/2
interface GigabitEthernet1/0/3
interface GigabitEthernet1/0/4
interface GigabitEthernet1/0/5
interface GigabitEthernet1/0/6
interface GigabitEthernet1/0/7
interface GigabitEthernet1/0/8
interface GigabitEthernet1/0/9
interface GigabitEthernet1/0/10
interface GigabitEthernet1/0/11
interface GigabitEthernet1/0/12
interface GigabitEthernet1/0/13
interface GigabitEthernet1/0/14
interface GigabitEthernet1/0/15
interface GigabitEthernet1/0/16
interface GigabitEthernet1/0/17
interface GigabitEthernet1/0/18
interface GigabitEthernet1/0/19
interface GigabitEthernet1/0/20
interface GigabitEthernet1/0/21
interface GigabitEthernet1/0/22
interface GigabitEthernet1/0/23
interface GigabitEthernet1/0/24
interface GigabitEthernet1/1/1
```

```
interface GigabitEthernet1/1/2
interface GigabitEthernet1/1/3
interface GigabitEthernet1/1/4
interface Vlan1
no ip address
router ospf 7
router-id 7.7.7.7
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ipv6 router ospf 7
control-plane
service-policy input system-cpp-policy
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
transport input ssh
line vty 5 15
login
transport input ssh
call-home
If contact email address in call-home is configured as sch-
smart-licensing@cisco.com
the email address configured in Cisco Smart License Portal will
be used as contact email address to send SCH notifications.
contact-email-addr sch-smart-licensing@cisco.com
profile "CiscoTAC-1"
active
destination transport-method http
end
```

Problems:

When configuring the second multi-layer switch we found that none of the ports were able to be put in the up state. We had already assigned ip addresses to the ports every port was in the down state. The switch was an older switch that was not updated when the other switches were. We could not figure out a solution to this issue and eventually just changed out the switch for a spare that had not been used yet. While we were able to get the

new switch to work, we had to redo the configurations as they were not fully documented (and they were incorrect). Many of our routers were also not updated to the most recent version of cisco IOS and were using one of the outdated routers. After successfully configuring IPv4 on the outdated router and the pings going through, we set up IPv6 which would not work despite what we thought was a correct configuration. After several hours of troubleshooting, we found the issue and switched to a router that was on the 16.9 version of Cisco IOS. Our rack was also not set up correctly and some of our console ports on the console hub were not in the correct devices and had to be moved to be ordered right to avoid confusion. Because of several issues, our successful traceroutes and pings were delayed.

Conclusion:

I learned that both single and multi-area OSPF can be viable in different situations. Multi-layer switches do have uses outside of just being both a layer 2 and 3 device and allow for some virtual routing on VLANs even if the switch wasn't needed in this specific lab. I also learned that things like versions can sometimes be the issue and should be checked along with configurations.